Locality of $\mathrm{Tr}(\phi^3)$ tree amplitudes from cyclic hidden zeros

Part II: the fat–zero generalisation

Abstract

Part I established locality through $n = 9$ by combining a single asymptotic "kill" on one hidden 1–zero with a per–n Laurent cascade. Here we promote the 1–zero to a *k–zero* ("fat zero"): the simultaneous imposition of the hidden–zero relations on k consecutive boundaries. A k–zero kills strictly more nonlocal coefficients than the 1–zero, with a clean geometric criterion: for every k consecutive edges, either a chord must cross the enclosing chord, or the $(k+1)$–gon spanned by those edges must extend to a $(k+2)$–gon built from chords of the diagram. The $(k+2)$–gons completing successive k–zeros build on top of each other — the $(k+2)$–gon at the k–zero is itself the $(k+1)$–gon at the next $(k+1)$–zero, so the family stacks. Triangulations always escape this test, and (conjecturally) only triangulations do. The Step–2 coefficient equations live in Part III.

1 From one zero to many

We keep the conventions of Part I. The n-gon has vertices $1, \ldots, n$; a chord X_{ij} ($1 \le i < j \le n$, $|j-i| \ne 1$, $(i,j) \ne (1,n)$) is a variable; an edge $X_{m,m+1}$ is a leg and equals 0. The ansatz is $B_n = \sum_{|M|=n-3} a_M / \prod_{c \in M} X_c$ over multisets M, and locality is the statement $a_M = 0$ whenever M contains a crossing pair.

A cyclic hidden 1*–zero* imposes the $n-3$ relations $c_{1j} = 0$, where

$$c_{ij} = X_{ij} + X_{i+1,\,j+1} - X_{i,\,j+1} - X_{i+1,\,j}. \tag{1}$$

On $\mathcal{Z}_{1,3}$ these read $X_{2k} = X_{1k} - X_{13}$ and $X_{2n} = -X_{13}$; the special chord is X_{13} and the bare chord is X_{2n}.

Definition 1 (k–zero)**.** For $1 \le k \le n-3$, the *k–zero based at* $\{1, \ldots, k{+}1\}$ is the simultaneous imposition

$$c_{ij} = 0, \qquad i = 1, \ldots, k, \quad j = k+2, \ldots, n-1. \tag{2}$$

It collects the hidden–zero relations on the k consecutive boundaries $(1,2), (2,3), \ldots, (k, k{+}1)$. For $k = 1$ it is the ordinary 1–zero. There are $k(n-k-2)$ relations and they cut out a linear surface $\mathcal{Z}^{(k)}$.

2 The constraint surface of a k–zero

Lemma 1 (Telescoping)**.** *On $\mathcal{Z}^{(k)}$, with the leg convention $X_{m,m+1} = 0$,*

$$X_{k+1,\,j} = X_{1j} - X_{1,k+2}, \qquad j = k+3, \ldots, n-1; \qquad X_{k+1,\,n} = -X_{1,k+2}, \tag{3}$$

$$X_{ij} = X_{k+1,\,j} + X_{i,\,k+2}, \qquad i = 2, \ldots, k, \quad j = k+3, \ldots, n. \tag{4}$$

Proof. Sum (1) over the full block of rows $i = 1, \ldots, k$ at fixed j. The two inner sums telescope, leaving $\sum_{i=1}^{k} c_{ij} = \big(X_{1j} - X_{k+1,j}\big) - \big(X_{1,j+1} - X_{k+1,j+1}\big) = 0$, so $X_{k+1,j} - X_{1j}$ is independent of j; evaluating at $j = k+2$ (with $X_{k+1,k+2} = 0$) gives (3), and at $j = n$ (with $X_{1n} = 0$) the bare relation. Summing the partial block $i' = i, \ldots, k$ gives $X_{ij} - X_{i,j+1} = X_{k+1,j} - X_{k+1,j+1}$; telescoping in j from $k+2$ yields (4). $\square$

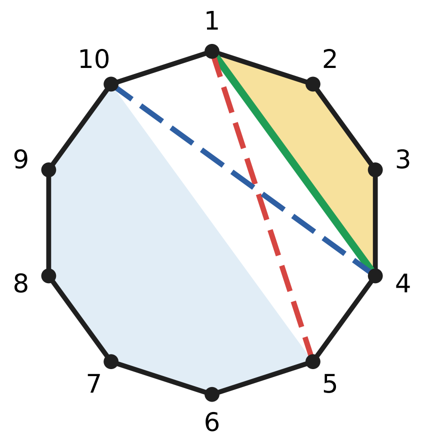

Figure 1: Geometry of a k–zero ($n = 10$, $k = 3$). Near block $N = \{1, 2, 3, 4\}$ (gold); far block $F = \{5, \dots, 10\}$ (blue). Green: the length–k enclosing chord $X_{1,4}$. Red dashed: the special $X_{1,5}$. Blue dashed: the bare $X_{4,10}$. Born–free chords are those internal to N or internal to F.

Three structures organise $\mathcal{Z}^{(k)}$ (Figure 1).

- The **near block** $N = \{1, \dots, k+1\}$ and the **far block** $F = \{k+2, \dots, n\}$. Every chord internal to N or to F is absent from (2): these are the *born–free* chords, untouched by the k–zero.
- The length–k **enclosing chord** $X_{1,k+1}$ closes the k consecutive boundaries into the $(k+1)$–gon N. It is itself born–free.
- The **special** $X_{1,k+2}$ and the **bare** $X_{k+1,n}$ — the exact analogues of X_{13} and X_{2n} at $k = 1$ — each bridge N to F across one boundary of N.

3 The geometric survival theorem

The kill mechanism is the Laurent argument of Part I: take an asymptotic limit on $\mathcal{Z}^{(k)}$ in which a chosen set of chords diverges; the terms whose denominators remain finite form a vanishing Laurent polynomial in the finite coordinates, forcing $a_M = 0$ for every M supported on the finite set. Running this argument with all the admissible limits leads to a single geometric criterion.

Theorem 1 (Survival on a k–zero)**.** *Fix $k \in \{1, \dots, n-3\}$ and a set of k consecutive polygon edges $(i, i+1), (i+1, i+2), \dots, (i+k-1, i+k)$, with associated enclosing chord $X_{i,i+k}$ and $(k+1)$–gon $N = \{i, \dots, i+k\}$. A multiset M survives the k–zero based at N iff at least one of the following holds:*

(P1) *M contains a chord that crosses the enclosing chord $X_{i,i+k}$ — equivalently, a chord with one endpoint strictly inside N and one strictly outside.*

(P2) *M extends N to a $(k+2)$–gon, i.e. M contains chords completing $\{i, \dots, i+k, v\}$ as a polygon for some far vertex $v \in F$.*

The argument is contained in note 22 page 2 and is the geometric shadow of the asymptotic limits on $\mathcal{Z}^{(k)}$: any term failing both (P1) and (P2) sits inside a strictly larger family of monomials that go to zero in the limit. We unpack (P2) explicitly in the next section.

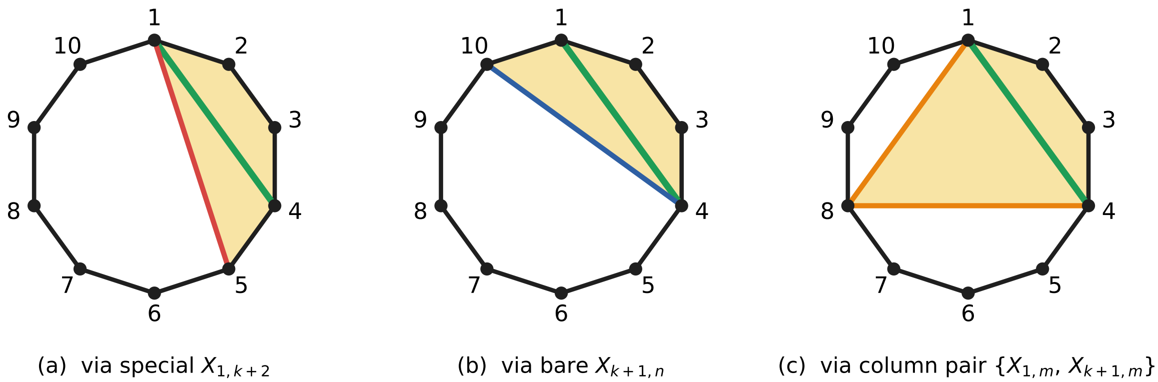

Figure 2: The three flavours of $(k+2)$–gon completion of the near block N ($n = 10$, $k = 3$, $m = 8$ for the column pair). For any non–crossing survivor, exactly one of these occurs.

4 P2: $(k+2)$–gons over $(k+1)$–gons

For a far vertex $v \in F = \{k+2, \ldots, n\}$, the $(k+2)$–gon $\{1, \ldots, k+1, v\}$ is closed in the polygon by:

(i) $v = k+2$: the edge $(k+1, k+2)$ together with the chord $X_{1,k+2}$ — the **special** closes it on the small side.

(ii) $v = n$: the edge $(n, 1)$ together with the chord $X_{k+1,n}$ — the **bare** closes it on the large side.

(iii) $v \in \{k+3, \ldots, n-1\}$: it takes *two* chords — $X_{1,v}$ and $X_{k+1,v}$ together, the **column pair**.

Stacking: each $(k+2)$–gon is the next $(k+1)$–gon

The completion is not a one–off geometric trick — it recurses. The $(k+2)$–gon that closes a survival at the k–zero is itself the $(k+1)$–gon at the $(k+1)$–zero based at the same starting vertex, which in turn must extend to a $(k+3)$–gon, and so on. So the family of k–zero survival conditions *stacks*: each step adds one more vertex to the enclosing polygon (Figure 3).

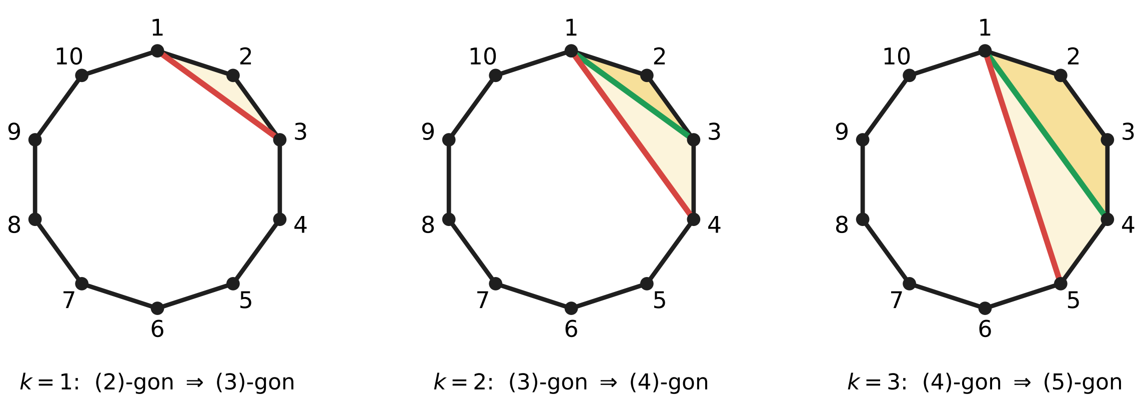

Figure 3: The stacking structure ($n = 10$). At k, the $(k+1)$–gon over k consecutive edges (darker shading) must extend to a $(k+2)$–gon (lighter shading) — and that $(k+2)$–gon is precisely the $(k+1)$–gon at the $(k+1)$–zero in the next step. The completions chain: triangle ⇒ 4–gon ⇒ 5–gon ⇒ $\cdots$.

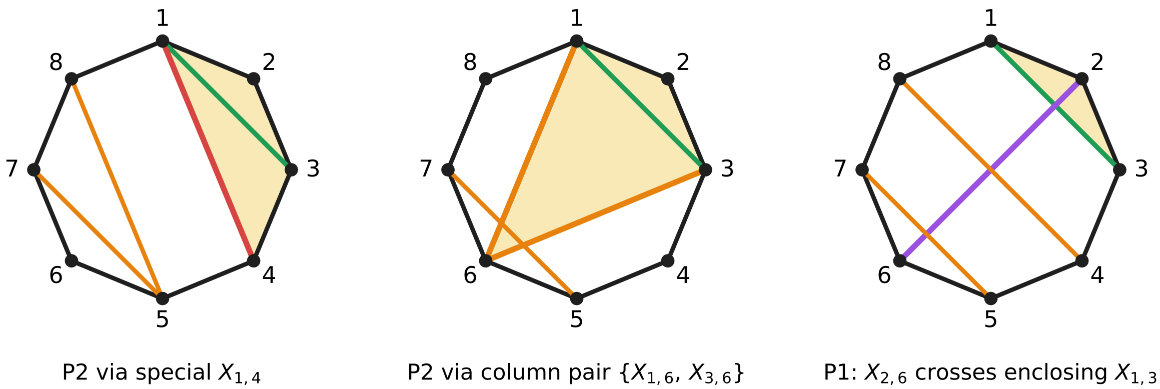

Figure 4: Survivors of the $k = 2$ zero based at $\{1, 2, 3\}$, $n = 8$. (a) and (b) are P2 completions of the triangle $\{1, 2, 3\}$ to a 4–gon — via the special $X_{1,4}$ and via the column pair $\{X_{1,6}, X_{3,6}\}$. (c) is P1: the chord $X_{2,6}$ crosses the enclosing chord $X_{1,3}$. (Whether (c) actually survives all k–zeros — not just this one — depends on whether its crossings cover all open regions.)

5 P1: crossing the enclosing chord

A multiset M can also survive by failing the no–crossing hypothesis at N — i.e. by including a chord that crosses the enclosing chord $X_{1,k+1}$. But surviving *every* k–zero this way is more subtle than a single crossing per k: the hidden–zero relations couple the boundary chords across the entire polygon, so a single crossing in one region typically still leaves another sub–polygon uncrossed (killing M via P2 there).

The multi–region rule. If a multiset's near complement breaks into several sub–polygons under crossing, *each* sub–polygon ("region") must carry its own crossing chord: *one cherry per region*. A single crossing isn't enough when the diagram has multiple open regions; survival requires multiple chords crossing *simultaneously*, one piercing each region (note 22 page 6).

Concretely, for a multiset M to survive a k–zero with $k \geq 2$ by P1, M must contain a chord set whose simultaneous crossings exhaust all the regions left open by the absent $(k+2)$–gon completion. A single crossing can cover only one region; configurations with two or more disjoint open regions therefore require two or more crossing chords at once.

6 Triangulations escape

A triangulation always satisfies the survival criterion: pick any k consecutive edges. If the $(k+1)$–gon N over those edges contains no chord crossing into the rest of the polygon, then adding the enclosing chord $X_{1,k+1}$ to the triangulation crosses nothing — so $X_{1,k+1}$ must already be a chord of the triangulation. Being a triangulation chord, it bounds a triangle on its far side, which extends N to a $(k+2)$–gon (P2). The conjecture of the program is the converse:

> **Conjecture.** Across all $k \in \{1, \ldots, n-3\}$ and all n cyclic rotations, the union of the k–zero survival conditions is satisfied exactly by the triangulations of the n–gon.

This is the geometric content of the all–n locality statement. Empirically through $n = 9$ the conjecture holds (Part I). Symbolically through $n = 8$ in the sympy test suite (`tests/stress_tests.py`, 17,602 non–crossing multisets enumerated, all matched), and end–to–end at $n \leq 7$ in the rank experiment of `computations/full_constraint_system/` (the full k–zero constraint matrix has cokernel exactly 1–dim, spanned by the triangulation indicator).

7 Status

Theorem 1 is the geometric kill criterion: for every k and every cyclic rotation, survival requires either a chord crossing the enclosing chord (P1) or a $(k+2)$–gon completion built from chords of the diagram (P2). The $(k+2)$–gons *stack* as k increases, so the family of conditions builds recursively, and triangulations satisfy all of them.

Establishing the converse uniformly in n — that *only* triangulations satisfy all k–zero conditions — requires reading the conditions together with the Step–2 coefficient equations, which fix the relative values of the surviving coefficients and rule out spurious survivors that satisfy the geometric criterion but not the algebraic identities. Those equations are classified in Part III.